Difference between Testis & Ovary

Aspect	Testis	Ovary
Location	Located in the scrotum, outside the body cavity	Located in the pelvic cavity, on either side of the uterus
Structure	Composed of seminiferous tubules, interstitial cells (Leydig cells), and Sertoli cells	Composed of ovarian follicles, corpus luteum, and stroma
Size	Generally larger than ovaries, about 4-5 cm in length	Smaller than testes, about 3-5 cm in length
Hormonal Regulation	Regulated by luteinizing hormone (LH) and follicle-stimulating hormone (FSH) from the pituitary gland	Regulated by LH and FSH, but also influenced by other hormones such as gonadotropin-releasing hormone (GnRH)
Gamete Production	Continuous production of sperm throughout life after puberty	Finite number of oocytes present at birth, with a monthly cycle of maturation and release during reproductive years
Developmental Stage	Starts producing sperm at puberty and continues throughout life	Oocyte development begins before birth, with maturation occurring during the menstrual cycle
Temperature Sensitivity	Requires a lower temperature for optimal sperm production, hence located outside the body	Functions optimally at body temperature
Function	Produces sperm and hormones (mainly testosterone)	Produces ova (eggs) and hormones (mainly estrogen and progesterone)

Difference between Trachea & Bronchii

Aspect	Trachea	Bronchii
Structure	Single, tubular structure that extends from the larynx to the primary bronchi.	Branching structures that arise from the trachea, leading to the lungs.
Diameter	Approximately 1-2 cm in diameter.	Decreases in diameter as they branch into secondary and tertiary bronchi.
Cartilage	Supported by C-shaped rings of hyaline cartilage.	Supported by irregular plates of cartilage that decrease in size with branching.
Epithelium	Lines with pseudostratified ciliated columnar epithelium.	Also lined with pseudostratified ciliated columnar epithelium, transitioning to simple columnar and cuboidal in smaller bronchi.
Location	Located anterior to the oesophagus in the neck and thorax	Located within the lungs, branching off from the trachea.
Muscle Layer	Contains a smooth muscle layer called the trachealis muscle at the back.	Contains smooth muscle that regulates airflow and bronchoconstriction.
Function	Conducts air to the bronchi and filters, warms, and moistens the air.	Conducts air into the lungs and participates in gas exchange at the alveolar level.

Difference between Artery & Vein

Aspect	Artery	Vein
Wall Structure	Thicker walls with three layers: tunica intima, tunica media (muscular and elastic), and tunica externa.	Thinner walls with three layers, but less muscular and elastic tissue compared to arteries.
Lumen Size	Narrower lumen to maintain high pressure of blood flow.	Wider lumen to accommodate lower pressure and larger volume of blood.
Color of Blood	Typically bright red due to oxygenated blood.	Typically dark red or bluish due to deoxygenated blood.
Location	Generally located deeper within the body.	Can be located both deep and superficial to the skin.
Valves	No valves present (except at the origin of major arteries).	Valves present to prevent backflow of blood.
Blood Pressure	High blood pressure due to proximity to the heart and the force of contraction.	Lower blood pressure as blood is returning to the heart.
Function	Carry oxygenated blood away from the heart to the tissues (except pulmonary arteries).	Carry deoxygenated blood back to the heart (except pulmonary veins).